CMSC 828N "Logical Aspects of Artificial Intelligence"

Homework 3 (Deadline: 0930am ET, Dec 2 2014)

We assume the existence of two disjoint sets of variables ***U*** and ***V***. ***V*** is the usual set of variables used to define terms and atoms.

An uncertain LP-rule has the form

$$A_0 : e_0 \leftarrow A_1 : e_1 \& \dots \& A_n : e_n$$

where $A_0, \dots, A_n$ are all atoms and the e_i's are all "u-expressions" which defined as follows.

- Every member of the unit real interval [0,1] is a u-expression.
- Any member of ***U*** is a u-expression.
- If u_1, u_2 are u-expressions, then $u_1 * u_2$ (denoting multiplication) and $u_1 \blacksquare u_2$ are *u-expressions.* When u_1, u_2 are ground u-expressions, $u_1 \blacksquare u_2 = u_1 + u_2 \; if \; u_1 + u_2 \leq 1 \; and \; 1 \; otherwise$.

An uncertain LP is a finite set of uncertain LP-rules.

All *u-expressions* can be evaluated to a value in the [0,1] interval as long as variable symbols have assignments in this range. An interpretation I is a mapping from the set of all ground atoms to the interval [0,1]. We can order the set of all interpretations as follows:
$I_1 \ll I_2 \; iff \; for \; all \; ground \; atoms \; A, I_1(A) \leq I_2(A)$.

An interpretation satisfies a ground uncertain LP-rule of the form shown above iff either $I(A_0) \geq e_0$ or there exists an $1 \leq i \leq n$ such that $I(A_i) < e_i$. Satisfaction of non-ground rules and an uncertain LP are defined in the usual way. You are required to (i) define an operator T_P associated with an uncertain LP P such that T_P is: (ii) monotonic and (iii) I is a model of P *iff* $T_P(I) \ll I$. Please include a proof of both assertions. In addition, state whether your definition of T_P is continuous or not – if it is, please provide a proof. Otherwise please provide a counter-example.

All class communication will be done via Piazza.